THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of the series $11 + 17 + 23 + \cdots + 113$ is: 1

(a) 1115 **(b)** 1118 **(c)** 1117 **(d)** 1116

2. How many terms of the AP $18, 15, 12, 9, \ldots$ give sum 0? 1

(a) 14 **(b)** 13 **(c)** 15 **(d)** 12

3. If the 3rd term of an AP is 8 and the 8th term is 3, then the 11th term is: 1

(a) 0 **(b)** 24 **(c)** 11 **(d)** 1

4. The sum of the first 15 terms of an AP is 465, and its first term is 10. The 15th term is: 1

(a) 52 **(b)** 53 **(c)** 54 **(d)** 51

5. In an AP, $S_{12} = 564$ and $S_{11} = 473$. Then $a_{12} =$ 1

(a) 90 **(b)** 92 **(c)** 91 **(d)** 93

6. The sum of the first 6 terms of an AP is $4n^2 + 5n$. Its common difference is: 1

(a) 9 **(b)** 10 **(c)** 7 **(d)** 8

7. The sum of the series $-8 + -5 + -2 + \cdots + 79$ is: 1

(a) 1066 **(b)** 1065 **(c)** 1064 **(d)** 1067

8. How many terms of the AP $30, 25, 20, 15, \ldots$ give sum 0? 1

(a) 14 **(b)** 13 **(c)** 15 **(d)** 12

9. If the 6th term of an AP is 7 and the 7th term is 6, then the 13th term is: 1

(a) 42 **(b)** 0 **(c)** 13 **(d)** 1

10. The sum of the first 11 terms of an AP is 495, and its first term is 5. The 11th term is: 1

(a) 86 **(b)** 87 **(c)** 85 **(d)** 84

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If $S_n = 2n^2 + 5n$, find the first term and the common difference. 2

12. In an AP, $a_3 + a_7 = 36$ and $a_2 + a_9 = 40$. Find a and d. 2

13. The 6th term of an AP is 11 and the 11th term is 6. Find the 17th term. 2

14. If $S_n = 4n^2 + 1n$, find the first term and the common difference. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an AP, $S_7 = 98$ and $S_{12} = 228$. Find the AP. 3

16. Find the sum of the first 14 terms of the AP whose 4th term is 21 and 9th term is 36. 3

17. Find the sum of all integers between 5 and 150 that leave remainder 1 when divided by 6. 3

OR

In an AP, $S_4 = 42$ and $S_8 = 164$. Find the AP. 3

18. Find the sum of the first 18 terms of the AP whose 3rd term is 18 and 6th term is 39. 3

OR

Find the sum of all integers between 30 and 400 that leave remainder 5 when divided by 11. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A ladder has 25 rungs whose lengths decrease uniformly from 70 cm at the bottom to 22 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

OR

A sum of ₹880 is to be given as 8 prizes, each prize ₹20 less than the preceding one. Find the value of the first and last prize. 5

20. A ladder has 19 rungs whose lengths decrease uniformly from 120 cm at the bottom to 48 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

OR

A sum of ₹1150 is to be given as 10 prizes, each prize ₹30 less than the preceding one. Find the value of the first and last prize. 5

END OF QUESTION PAPER — Total: 40 Marks

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